

Summary Report for Azure Free Tier Setup and Cloud Deployment Exercise

1. Introduction

This report summarizes the process of creating and configuring a Microsoft Azure Free Tier account, deploying basic cloud management resources, and applying core cloud governance principles. The exercise focused on account setup, portal navigation, cost control configuration, resource organization, and security awareness.

2. Selected Azure Region

During the setup of resources in Microsoft Azure, the chosen region for deploying resources was **South Africa North**.

Reason for Selection:

- It is the closest Azure region to Nigeria, ensuring lower network latency.
- It provides better performance for applications accessed from West Africa.
- It supports most core Azure services required for development and testing.
- It helps maintain compliance with data residency and regional data handling preferences.

Where South Africa North was not available for specific services, **West Europe** was considered as a fallback option due to its stability and availability of services.

3. Overview of Azure Free Tier Setup

The Azure Free Tier account was successfully created using a Microsoft account with identity verification via email and phone number.

Key Components Activated:

- Azure Free Account with \$200 credit for 30 days
- Access to selected free services for 12 months
- Access to always-free services with usage limits
- Azure Portal access for cloud resource management

4. Resource Deployment and Configuration

4.1 Resource Group Creation

A resource group named **3MTT-Lab-RG** was created to logically organize and manage cloud resources.

Resource groups in Azure serve as containers that group related resources for easier management, monitoring, and cost tracking.

4.2 Cost Management and Budget Setup

A budget named **FreeTierLimit** was configured under Azure Cost Management.

Configuration Details:

- Budget scope: Subscription level
- Purpose: Prevent accidental overspending beyond free-tier allocation
- Alert threshold: 50% of allocated usage
- Monitoring enabled for cost tracking

This ensures financial governance and aligns with cloud cost management best practices.

5. Portal Navigation and Resource Management

The Azure Portal was used as the primary interface for managing cloud resources.

Key Navigation Elements Used:

- Dashboard for overview of resources
- Search bar for quick access to services
- Resource Groups section for resource organization
- Cost Management + Billing section for financial tracking
- Settings panel for account configuration

The portal provides a centralized management interface for all Azure services.

6. Security Considerations

Security was considered an essential part of the setup process.

Security Measures Observed:

- Microsoft account authentication secured access to Azure portal
- Password protection enforced via Microsoft identity system

- Multi-Factor Authentication (MFA) recommended for enhanced security
- Azure security recommendations available through the Security Center

Best Practices Identified:

- Enable MFA for all cloud accounts
- Use strong, unique passwords
- Regularly monitor account activity and billing alerts

7. Free Tier Limitations and Benefits

The Azure Free Tier provides the following benefits:

Benefits:

- \$200 credit valid for 30 days
- Free access to selected services for 12 months
- Always-free services with limited usage thresholds

Limitations:

- Credit expires after 30 days if unused
- Some services automatically convert to pay-as-you-go after free period
- Usage limits apply to prevent overspending

Understanding these limits is essential for proper cost control and resource planning.

8. Shared Responsibility Model (Explanation)

The Microsoft Azure Shared Responsibility Model defines how security and management responsibilities are divided between Microsoft (the cloud provider) and the customer (the user, in this instance – Victor Mgbah).

Microsoft's Responsibility (Cloud Provider)

Microsoft is responsible for:

- Physical security of data centers
- Hardware infrastructure (servers, storage, networking)
- Virtualization layer and hypervisor security

- Ensuring availability and redundancy of cloud services
- Maintaining Azure platform updates and patching of underlying systems

Customer Responsibility (User / Victor Mgbob)

As the Azure user, the responsibilities include:

- Configuring and managing virtual machines or storage accounts
- Securing access using identity and access management (IAM / Entra ID)
- Applying security patches inside the operating system (for VMs)
- Managing network security rules (firewalls, NSGs)
- Monitoring usage and cost through Azure Cost Management
- Ensuring proper deletion of unused resources to avoid charges

9. Conclusion

The Azure Free Tier setup exercise successfully introduced core cloud computing concepts including resource management, cost control, portal navigation, and security governance.

The configured environment demonstrates basic competency in cloud onboarding, with proper implementation of:

- Resource grouping
- Budget control
- Portal navigation
- Security awareness
- Region-based deployment considerations

This foundational setup provides a base for more advanced cloud computing tasks such as virtual machine deployment, networking configuration, and application hosting.